

# Soft Water, Strong Odor



Larry and Suzanne Weingarten operate Elemental Enterprises in Monterey, California, which services conventional and solar water heaters.

**Q. I have a gas-fueled direct-vent hot water tank. It has been in service since September 2003. I have a water softener as well. We get our water from a well. Before being softened, the water has a hardness reading of 33 grains. My hot water has a reading of 1 grain of hardness after softening. My cold water is 0 grains. Before I went away for 13 days, I set the temperature to the lowest setting. Upon my return, the hydrogen sulfide smell could knock you out! Finally, the smell has almost entirely dissipated. I have some questions:**

**I understand that you can replace the standard magnesium anode with an aluminum anode and not get the bad smell. Are there any drawbacks to going to aluminum?**

**Is it correct that the anode rod needs to be replaced every 1 to 2 years?**

**If I am going to be gone for an extended period of time, can I just turn off the gas to the tank till I get back? The tank is located in the basement. The winters do bring below freezing temperatures. Should I partially drain the tank?**

**Is there any way to get the hardness to 0 in the tank? It won't go below 1 because of buildup. Any ideas?**

**—No More Rotten Eggs**

**A.** Step slowly back from the softener with that bag of salt! Your water is softened too much, and this is certainly contributing to the smelly water problem. Softening exchanges sodium for the calcium and magnesium in water.

Sodium is more conductive, so it speeds up the corroding of the anode. Note that an anode is put in every glass-lined water heater to prevent rusting of the tank. Oversoftening greatly accelerates anode consumption. Normally an anode will last four to six years. In over-softened water, an anode may last only six months.

As anodes corrode, hydrogen gas is generated in the tank. Some anaerobic bacteria like hydrogen and warmth. That rotten-egg odor is their gift to you. An oversoftened water heater gives them the perfect home with lots of hydrogen gas. The National Association of Corrosion Engineers suggests leaving 60 to 120 parts per million (ppm) of hardness in water after it has been softened. One grain is about 18 ppm.

Your water is 33 grains hardness, or about 594 ppm. Taking all that hardness out means that you're putting a lot of salt back in. As a first step, try leaving 6 grains hardness in the cold water. That will help the anode in your heater to last, indirectly helping the heater to live longer. It will also allow a film of scale to build up inside any copper piping in your house, which will help prevent copper corrosion. Blue or green staining at fixtures lets you know that the copper piping is being attacked. Also, that level of softening will reduce the amount of hydrogen gas in the tank, making it less hospitable to the bacteria.

Whenever a house goes unoccupied, the heater is much more prone to developing odor. Lack of use gives the bacteria a chance to flourish in the tank, since the tank is not being constantly rinsed out. When you return home after a week or more, it might be a good idea to treat the water with hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>). The bacteria don't enjoy the additional oxygen that hydrogen peroxide brings. Use the standard 3% solution found in supermarkets. One or 2 pints per 40 gallons of capacity usually works to get rid of the odor, and steady water use thereafter will keep the odor gone. Put the peroxide in the tank, possibly through the anode port—this takes a bit of plumbing skill, but you have to

open the anode port to check or replace the anode anyway—and then run each hot tap just until it warms. This way you are treating all of the affected plumbing.

If you prefer not to open the anode port for putting in the peroxide, use the hot water connection, unless there is a heat trap, which makes it difficult to pour in the hydrogen peroxide. There is a trick that Ontario Hydro used to suggest. After the water is shut off to the heater (and the heat source is off), put a hose on the drain valve and open it, then open a hot tap. Air should be pulled in. If so, a pan of peroxide can be held so that it is pulled into the faucet and hopefully to the tank. That method might be good for those plumbing-averse readers.

When possible, magnesium anodes are preferable to aluminum, since magnesium is safer to ingest and produces far less corrosion byproduct. Aluminum anodes can make gas heaters noisy with popping and rumbling sounds in as little as three months, because of the sediment they produce. For difficult odor problems there is an anode made of aluminum with 8% zinc. Another anode to use is magnesium with a resistor installed at the top. Rheem or GE heaters use these. The resistor slows the anode down, generating less hydrogen.

It is safest not to drink or cook with water that's been in the heater, particularly if you don't know what type of anode it has. You will probably need to replace the anode in your tank soon. I'd try a resistor type first. With softened water, check the new anode in two years to see how it is doing. It needs replacing when 6 inches of its core wire is exposed. You'll be able to judge replacement intervals by checking the wear on the rods.

Freezing is a tricky question. If you drain down the tank without completely drying the inside, it will rust. When it's full, the anode is preventing damage. You could use a small lightbulb near the tank or heat tape on it to keep the tank from freezing. Or consider simply leaving the pilot on.

